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SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

"From individual interaction to shared, continuously improving skill ecosystems"

PROBLEM

Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

⚠ Static Skill Ecosystem (Today)

- Skills manually installed from centralized hub
- No updates from runtime interaction or experience
- Failures and solutions are siloed per session
- Each user rediscovers the same patterns independently
- No system-level knowledge accumulation

Memory-based methods: tied to specific instances
Skill-based methods: static library, never evolves

✓ SkillClaw — Evolving Ecosystem

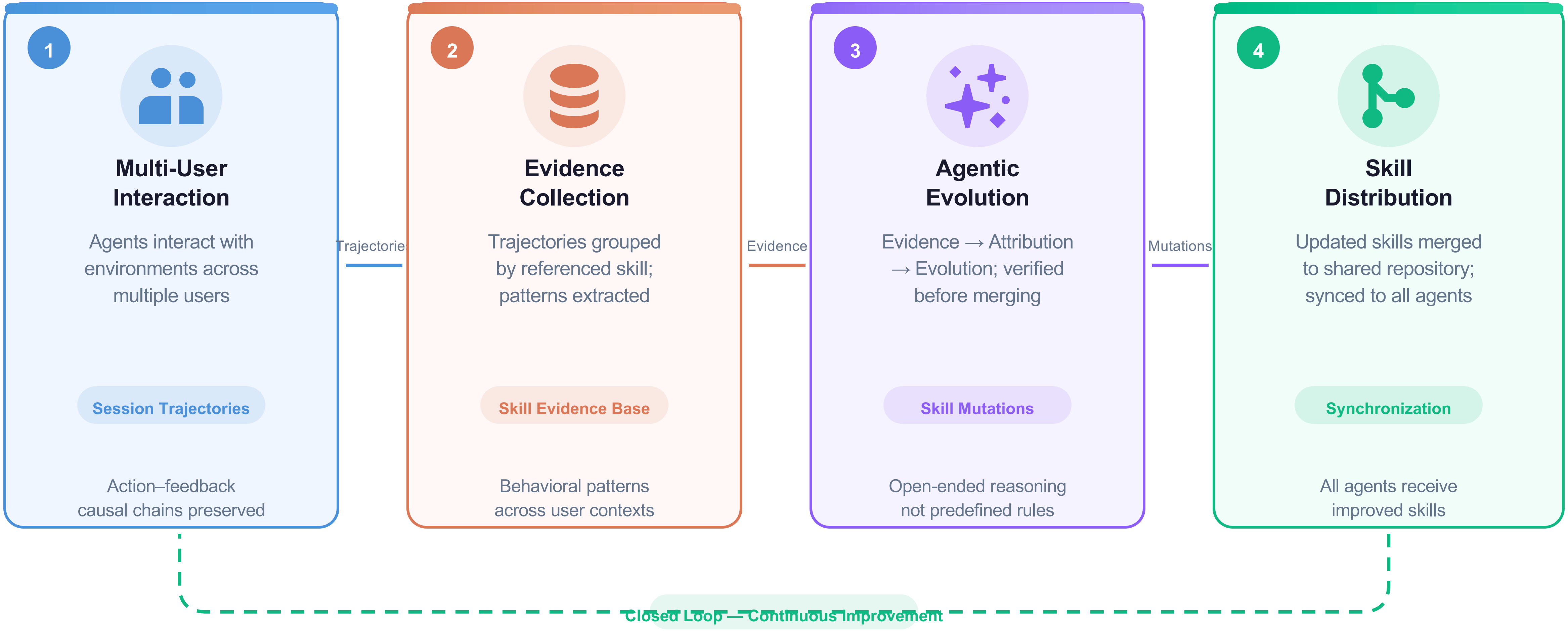
- Skills evolve continuously from runtime usage
- Cross-user interaction signals aggregated automatically
- Successful solutions persist and propagate system-wide
- Knowledge compounds with every interaction
- Agentic evolver reasons over evidence — no rules

Closed-loop: Interaction → Evolution → Distribution
Fully automatic — no manual curation required

"Same failure patterns are rediscovered by every user independently — the system never learns"

ARCHITECTURE

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

• Stage 1: Evidence → • Stage 2: Attribution → • Stage 3: Evolution

Stage 1
Evidence

- Analyze cross-user trajectories
- Identify behavioral patterns
- Group by referenced skill
- Separate success from failure contexts across diverse users

Input: Aggregated session
trajectories grouped by skill

Stage 2
Attribution

- Diagnose root causes of failures
- Identify success conditions
- Pinpoint skill behaviors that cause problems across contexts
- Distinguish generalizable issues from idiosyncratic edge cases

Output: Root cause diagnosis
of skill behavioral boundaries

Stage 3
Evolution

- Propose skill mutations via open-ended agent reasoning
- Refine existing skills or create entirely new ones
- Verify before merging into shared repository

Not predefined rules —
autonomous reasoning over evidence

Evolution is driven by an autonomous agent performing open-ended reasoning over interaction evidence — not predefined update rules

CROSS-USER SIGNALS

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Why Single-User Signals Are Insufficient

- A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix
- Sessions expose only a narrow slice of a skill's behavioral boundary — one context, one perspective
- No system-level signal to distinguish real regression from context-specific behavior

What Cross-User Aggregation Provides

- Complementary views of each skill's behavioral boundary
- Reveals both conditions where skills work and those where they break — across diverse contexts
- Different users exercising the same skill in diverse contexts produce complementary signal

Compatible Claw-Style Agent Systems

OpenClaw
Base ecosystem

CoPaw
Cooperative variant

IronClaw
Robust variant

PicoClaw
Lightweight variant

ZeroClaw
Zero-shot variant

NanoClaw
Nano-scale variant

NemoClaw
Enterprise variant

7 compatible systems — each contributes unique interaction data

Diverse contexts → Complementary failure & success signals → Generalizable skill improvements

WildClawBench evaluation with Qwen3-Max demonstrates substantial improvements across tasks

CASE STUDY 1

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

1 of 6 candidates accepted

Conservative verification prevents regressions — the evolver rejects when evidence does not clearly support improvement

Day	Candidate Skill	Outcome	Rationale / Next-Day Action
Day 1	**validate-tmp-workspace-inputs** Check /tmp_workspace inputs & environment setup	✓ ACCEPTED	Upgrade to new best pool Baseline improvement confirmed by validator
Day 2	**multimodal-input-validation-and-setup** Multimodal input validation & output env init	✗ REJECTED	Keep current best pool No net improvement over current best pool
Day 3	**multimodal-creative-task-pipeline** Multimodal creative pipeline (extract from PDF/video/image)	✗ REJECTED	Keep current best pool No improvement over pool
Day 4	**multimodal-creative-task-pipeline (impr.)** Added image classification, visual generation, structured validation	✗ REJECTED	Keep current best pool Still no improvement over pool
Day 5	**multimodal-creative-task-pipeline + validate-required-input-files** Creative pipeline + per-file fail-fast validation	✗ REJECTED	Keep current best pool No improvement over current pool
Day 6	**multimodal-creative-task-pipeline (cand.)** Extended PDF-to-poster / document-to-visual paths	✗ REJECTED	Not admitted to next cycle Cycle ends; skill pool remains conservative

1 Accepted | 5 Rejected | Conservative gating: only Day 1 baseline exceeded the current best pool threshold

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

3 of 6 candidates accepted — successful iterative refinement before reaching a plateau

Contrasts with creative pipeline case: evidence-supported improvements are adopted; plateau is a sign of maturity, not failure

Day	Change Summary	Outcome	Rationale / Next-Day Action
Day 1	**git-push-with-auth-fallback** Safe fallback instead of blocking on push failure	✓ ACCEPTED	Add to Safety best pool Clear behavioral improvement confirmed
Day 2	**git-push-with-auth-fallback** Unified patch/bundle filenames, reduced inconsistency	✓ ACCEPTED	Keep updated best pool Reduces naming inconsistencies
Day 3	**git-push-with-auth-fallback + git-clone-to-directory** Auth-alternative paths & secrets audit; fixed subshell pitfalls	✓ ACCEPTED	Keep current best pool Addresses auth edge cases
Day 4	**(none — same-pool retest)** Validator confirmed current pool as best; no new candidate	✓ ACCEPTED	Continue same best pool Stability confirmed — pool is optimal
Day 5	**git-push-with-auth-fallback** Added "push hang treated as auth failure"; no improvement	✗ REJECTED	Keep current best pool No improvement over current best pool
Day 6	**git-push-with-auth-fallback** Added identity config & filename consistency; no improvement	✗ REJECTED	Not admitted to next cycle Plateau reached — skill has matured

3 Accepted · 2 Rejected · Iterative refinement succeeds when evidence clearly supports improvement; plateau shows skill maturity

KEY TAKEAWAYS

Three Properties That Distinguish SkillClaw from Existing Systems

1

Collective Evolution

Individual interactions contribute to a shared, continuously improving skill ecosystem

Knowledge accumulates at
the system level — not per user

2

Fully Automatic

Skill evolution is driven by runtime interaction — no manual curation or user intervention

Zero human curation required
in the evolution pipeline

3

Agentic Evolution Paradigm

Skill updates produced through open-ended agent reasoning over evidence — not rules

Context-aware mutations via
autonomous agent reasoning

Implication for LLM Agent System Design

Build systems where every user interaction contributes to skill improvement across the ecosystem

Evaluation on WildClawBench (Qwen3-Max) demonstrates substantial improvements